F18CF – Linear Algebra

Tutorial Sheet 6: Rank, Nullity, and Isomorphisms

Getting started.

1. For each of the following matrices, determine the rank, the nullity, a basis of the kernel and a basis of the image:

$$(i) \quad \begin{pmatrix} 1 & -1 & 2 & 1 \\ 2 & 1 & -1 & 2 \\ 2 & -1 & 3 & -1 \\ 1 & 1 & -2 & 4 \end{pmatrix}, \qquad (ii) \quad \begin{pmatrix} 1 & 2 & -1 & 4 \\ 2 & -1 & 3 & 3 \\ -1 & 3 & -4 & 1 \\ 0 & 1 & -1 & 1 \end{pmatrix}.$$

2. Let A be a 3×4 matrix with rk(A) $= 3$.

   (a) What is the dimension of $\ker(A)$?

   (b) Is the equation $A\vec{x} = \vec{b}$ consistent for all $\vec{b} \in \mathbb{R}^3$?

   (c) If the equation $A\vec{x} = \vec{b}$ is consistent, how many solutions does it have ?

   If instead A is a 4×3 matrix with rk(A) $= 3$ and we consider the equation $A\vec{x} = \vec{b}$ for $\vec{b} \in \mathbb{R}^4$, what are the answers to (a), (b), (c) ?

3. We consider the linear map

$$T : \mathbb{R}^3 \to \mathbb{R}^3, \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} x + y + z \\ x + 2y + 4z \\ x + 4y + 16z \end{pmatrix}.$$

   (a) What is the matrix of T in the standard basis of $\mathbb{R}^3$?

   (b) What is the matrix of T in the basis $(\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix})$ of $\mathbb{R}^3$?

Problems.

4. **An image problem.** The goal of this problem is to illustrate how the Rank-Nullity Theorem can be used to describe certain subspaces.

   We consider the map

$$T : \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_n(\mathbb{R}), \quad P(x) \mapsto P(x+1) - P(x).$$

   (a) Show that T is a linear map.

   (b) Compute the kernel of T. What is its dimension?

(c) What is the dimension of $\mathrm{im}(T)$?

(d) Deduce that $\mathrm{im}(T) = \mathcal{P}_{n-1}(\mathbb{R})$.

5. **Magic squares.** A 3×3 *homogeneous magic square* is a 3×3 matrix such that the sum of the coefficients in each row and each column is 0.

(a) Show that the set W of all homogeneous magic squares is a subspace of $M_3(\mathbb{R})$.

(b) We introduce the map

$$T : W \to \mathbb{R}^4, A \mapsto (a_{11}, a_{12}, a_{21}, a_{22})^T.$$

Show that T is a linear map.

(c) Show that T is an isomorphism. What is the dimension of W?

6. **Sums of consecutive powers.** The goal of this problem is to use linear algebra methods to find formulas for the following sums: $\sum_{k=1}^{n} k, \quad \sum_{k=1}^{n} k^2, \quad \sum_{k=1}^{n} k^3$.

We introduce the polynomial functions $P_0(x) := 1, \quad P_1(x) := x$, and for $n \geq 2$,

$$P_n(x) := \frac{x(x-1)\cdots(x-n+1)}{n!}.$$

We work in the vector space $\mathcal{P}_3(\mathbb{R})$ with its standard basis $\mathcal{B} = (1, x, x^2, x^3)$.

(a) Show that $\mathcal{B}' := (P_0(x), P_1(x), P_2(x), P_3(x))$ is also a basis of $\mathcal{P}_3(\mathbb{R})$.

(b) Compute the matrix of change of basis $[\mathcal{B}']_{\mathcal{B}}$, and compute the matrix of change of basis $[\mathcal{B}]_{\mathcal{B}'}$.

(c) Show that, for $0 \leq i \leq 3$, we have

$$P_{i+1}(x+1) - P_{i+1}(x) = P_i(x),$$

and deduce that for $n \geq 1$, we have

$$\sum_{k=0}^{n} P_i(k) = P_{i+1}(n+1).$$

(d) Use questions (b) and (c) to find a formula for $\sum_{k=1}^{n} k$, $\sum_{k=1}^{n} k^2$, and $\sum_{k=1}^{n} k^3$ as a polynomial expression in n.